

Design and Analysis of a Novel Axial-Flux Slotless Limited-Angle Torque Motor With Trapezoidal Cross Section for the Stator

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Abstract—This paper presents a novel limited-angle torque motor as a modified double-sided slotless axial-flux permanent-magnet motor (DSAPM) with a trapezoidal cross section for the stator. A new approach to the magnetic equivalent circuit and semi-three-dimensional (3-D) analytical armature reaction field modeling for the proposed actuator are derived and implemented in the design process. Designing is done based on avoidance of saturation at maximum torque production to prevent distortion of torque characteristic in short-time operations. Next, finite-element analysis is utilized to confirm the accuracy of the design equations and to compare the proposed actuator with conventional industrial limited-angle torque motors and the conventional DSAPM structure. Finally, the proposed actuator is prototyped and its experimental results are presented and discussed. Finite-element analysis and experimental data show better accuracy of the model, electromagnetic capability improvements by this new structure and excellent applicability of the proposed rotary actuator.

Index Terms—Armature reaction, axial-flux permanent-magnet (PM) machines, field calculation, finite-element analysis (FEM), limited-angle torque motor (LATM), magnetic equivalent circuit (MEC), Maxwell's equations, rotary actuator, saturation avoidance, slotless machines.

I. INTRODUCTION

ELECTROMECHANICAL rotary actuators that have a limited movement generally smaller than $\pm 180^\circ$ without any mechanical torque conversion parts such as the gearbox have been called limited-angle torque motors (LATMs). Basic topology of different types of LATMs is similar to the conventional rotating machines. However, LATMs have a uniform torque profile during operation range, constant torque to input current ratio, higher torque density compared to the conventional rotating machines, and fewer mechanical parts. These features will result in simple and accurate position controlling, smaller volume, and lower maintenance compared to any electromechanical and mechanical position controlling systems.

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Slotted and slotless LATMs are widely used in position controlling systems from simple on-off valves to accurate tracking of a reference signal [1]. The slotted types have more torque density due to larger magnetic loading, but the slotless types are more common in the industries. This is the result of cogging torque production in salient and slotted structures, which is not acceptable in accurate control utilization of the LATMs [1]. Simple structure but accurate controlling and high reliability make them the best choice for servo-valve actuators that require torques under 7 N·m [2]. Controllable fuel and exhaust valves in combustion engines and gas turbines are the most remarkable applications of LATMs [1], [3], [4]. They are used in accurate robotic arms that are applied widely in electronic device manufacturing, medical robots, precise scanners, etc. [5], [6].

Already, many different structures of LATMs and their performances have been introduced and discussed. The laws-relay actuator is similar to the radial flux-reversal machine that was introduced and analyzed as a narrow-angle constant torque LATM [2]. Electromechanical and thermal analysis of wide-angle doubly excited doubly salient concentrated-winding LATMs were analyzed in [7] and [8]. The two previous actuators have been used as high torque-density actuators. However, due to nonuniform output torque especially at high excitations, neither became industrially manufactured in general. Radial flux permanent magnet LATMs are similar to the radial flux BLDCs [9].

Two common types of axial flux LATMs have been introduced and widely utilized in the industry until now. Voice coil motor (VCM) has the coreless structure with PMs as fixed parts and armature coil as a rotating section [10], [11]. Due to low-inertia rotor and uniform torque-angle profile, their main application is in a precise and high-speed position-controlling system such as hard disks and precise scanners [11]. Another LATM is a pancake type similar to the pancake type BLDC that has a slotted stator with concentrated armature windings [12]. Main utilization of the pancake LATMs is in controlling fuel and exhaust valves in vehicle combustion engines [4]. They are used as high torque-density LATMs. Due to slotted stator, pancake types have dome-shape torque-angle profile, which could make position controlling more complicated than for the slotless LATMs with constant torque-angle profile [4]. Hence, no precise position controlling applications for the slotted types have been issued until now.

In this paper, as shown in Fig. 1(a), a primary studied LATM is a slotless double-sided axial-flux actuator with the rectangular stator cross section (DAAR). This is similar to the conventional double-sided slotless axial-flux PM motors with

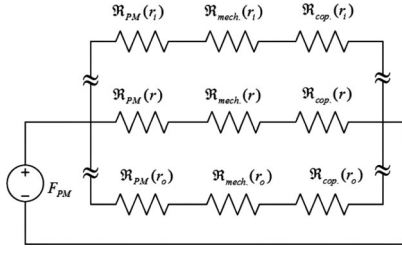


Fig. 4. MEC of a DAAT.

A. Magnetic Equivalent Circuit

Here, a special MEC for a DAAT is proposed. It has continuously varying elements as functions of r unlike the ordinary ones with discrete elements. Magnetic characteristic of the excitation is analyzed by the proposed MEC with the following simplifications:

- 1) The mmf drop in the ferromagnetic parts is neglected (the rotors and the stator).
- 2) Edge effects in the inner and outer radii of the actuator are ignored.
- 3) The rotor is positioned such that its direct axis is aligned with the quadrature axis of the stator (no armature reaction field effect on the excitation).
- 4) Linear characteristic of the PM is utilized.

Based on the earlier assumptions, the proposed MEC for DAAT is shown in Fig. 4.

The symbols are defined as the following:

- 1) The mmf of the PM:

$$F_{PM} = H_c l_m \quad (3)$$

- 2) The reluctance of a differential section of PM at distance r away from the center of the actuator:

$$\mathfrak{R}_{PM}(r) = l_m [\mu_m A_m(r)]^{-1} \quad (4)$$

- 3) The reluctance of mechanical clearance at radius r :

$$\mathfrak{R}_{mech.}(r) = l_{mech} [\mu_0 A_g(r)]^{-1} \quad (5)$$

- 4) The reluctance of the winding at radius r

$$\mathfrak{R}_{cop.}(r) = h_{cop.}(r) [\mu_0 A_g(r)]^{-1} \quad (6)$$

where H_c , l_m , l_{mech} , μ_m , and μ_0 are, respectively, coercive force of the PM, length of the PM, length of the mechanical clearance, the permeability of the PM, and the permeability of the free space.

At radius r , the magnetic flux density of the air gap under each permanent magnet $B_g(r)$ can be calculated according to the proposed MEC and (1) as the following:

$$\begin{aligned} B_g(r) &= \frac{H_c l_m}{[r_o h_{cop.o} r^{-1} + l_{mech}] \mu_0^{-1} + l_m \mu_m^{-1} s^{-1}} \\ &= \frac{a_1 r}{a_2 + a_3 r} \quad |_{0 < s \leq 1} \end{aligned} \quad (7)$$

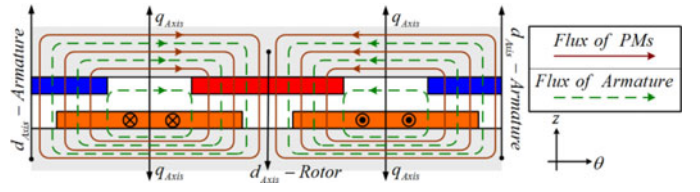


Fig. 5. Magnetic flux lines in the critical position of the rotor (one lateral scheme).

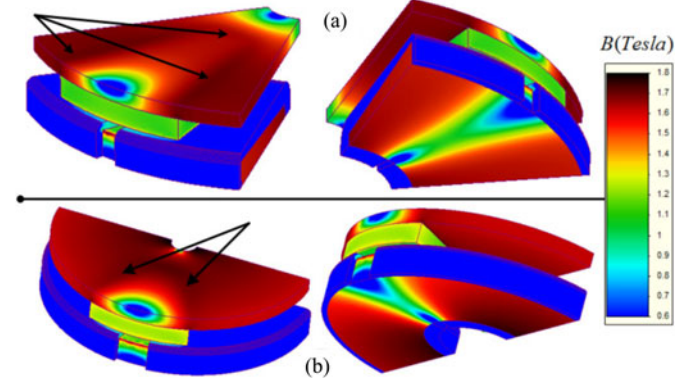


Fig. 6. 3-D FEA: (a) magnetic flux density distribution of the DAAT-20 and (b) magnetic flux density distribution of the DAAT-120.

where s is the fringing effect coefficient and also a_1 , a_2 , and a_3 are defined by the following expressions:

$$a_1 = H_c l_m \quad (8)$$

$$a_2 = r_o h_{cop.o} \mu_0^{-1} \quad (9)$$

$$a_3 = l_{mech} (\mu_0)^{-1} + l_m (\mu_m s)^{-1} \quad (10)$$

B. Magnetic Flux Density Calculation in Ferromagnetic Parts

In this paper, the design concept for DAAT is based on the effective saturation avoidance in order to maintain its linear characteristic. The actuator is supposed to produce a maximum output torque without distortion (almost completely flat) in the constant-torque range for a short-time operation.

As seen in Fig. 5, if the d -axes of the rotors and the stator are away from each other by $2\pi.p^{-1}$ radians, flux lines of the PMs and the armature windings have the same direction and congest on the q -axis of the rotors. This position is considered as the critical operating point in the design geometry of DAAT. If saturation is avoided in this position, one can be certain that linear behavior of the actuator is maintained in other operating points too.

Furthermore, if the inner radii of the rotor and the stator are not the same, another critical area will exist in the rotor, which is illustrated by the arrows in Fig. 6. This phenomenon is neglected in the analytical design. If it causes the nonlinear behavior of the actuator in the FE analysis, a larger value for the rotor thickness should be chosen to relieve the distortion in the torque profile.

By avoiding saturation in the design procedure, the material of the analyzed device could be assumed linear. So, fluxes produced by the excitation and the armature windings can separately be calculated as presented in the following subsections.

1) *PMs Magnetic Flux*: Using MEC, the differential magnetic flux in the air gap $d\phi_g$ passing through the differential surface element at radius r with the air gap angle θ_g is defined as

$$d\phi_g(r) = \theta_g B_g(r) r dr \quad (11)$$

Now, the total magnetic flux $\phi_{T.g}$ can be obtained by using (7) and (11) as follows:

$$\phi_{T.g} = \int_{r_i}^{r_o} d\phi_g(r) dr = \theta_g a_1 a_3^{-3} \left[\frac{(a_3 r)^2}{2} + a_2^2 \ln(a_2 + a_3 r) - a_2 a_3 r \right] \Big|_{r: r_i \rightarrow r_o}. \quad (12)$$

2) *Magnetic Flux of the Armature*: In this section, magnetic flux produced by the armature current is calculated by a semi-3-D method. Since the stator windings are toroidal, the armature flux can be separated into two different parts as:

- 1) flux passing through the inner and the outer walls of the stator (caused by the axial armature current);
- 2) flux passing through the lateral surfaces of the stator (caused by the radial armature current).

The magnetic flux produced by the stator windings due to the axial armature current next to the outer or the inner walls of the stator enters one half of the winding, passes through the q -axis of the armature, then goes out of the other half, and finally closes its path through the free space. The description and calculation of this magnetic flux are in Appendix A. Now, the total fluxes of windings passing over the armature q -axis due to the inner and outer surfaces of the stator are, respectively, defined by equations (13) and (14).

$$\begin{aligned} \phi_{s.in} &= \int_0^{\frac{\pi}{p}} B_{s.in}(r = r_i, \theta) r_i d_i d\theta = \\ &\begin{cases} \sum_{n=2k+1} \left[E_n r_i^{\frac{np}{2}} + F_n r_i^{-\frac{np}{2}} + r_i^2 \alpha_n \right] d_i, & np \neq 4 \\ \sum_{n=2k+1} \left[E_n r_i^{\frac{np}{2}} + F_n r_i^{-\frac{np}{2}} + \alpha_n r_i^2 \ln(r_i) \right] d_i, & np = 4 \end{cases} \end{aligned} \quad (13)$$

where F_n , E_n , and α_n are, respectively, defined by (A.12), (A.13), and (A.15) in the Appendix A.

$$\begin{aligned} \phi_{s.out} &= \int_0^{\frac{\pi}{p}} B_{s.out}(r = r_o, \theta) r_o d_o d\theta = \\ &\begin{cases} \sum_{n=2k+1} \left[I_n r_o^{\frac{np}{2}} + J_n r_o^{-\frac{np}{2}} + r_o^2 \alpha_n \right] d_o, & np \neq 4 \\ \sum_{n=2k+1} \left[I_n r_o^{\frac{np}{2}} + J_n r_o^{-\frac{np}{2}} + \alpha_n r_o^2 \ln(r_o) \right] d_o, & np = 4 \end{cases} \end{aligned} \quad (14)$$

where I_n and J_n are defined by (A.17) and (A.18), respectively.

The magnetic flux produced by the stator windings due to the radial armature current next to the lateral surfaces of the stator enters one half of the winding, passes in the q -axis direction of the armature, exits from the other half of the winding, and

finally closes its path through the rotors. To avoid complexity in calculations due to the nonuniform nonmagnetic gaps, the armature winding and its adjacent air gap are divided into i thickness bands, each with a thickness m . By summing up the calculated magnetic fluxes in each band, the total flux by the lateral winding surfaces is obtained.

To obtain flux distribution, Maxwell's equations in terms of magnetic vector potential in the angular cross section of the stator are solved as in the Appendix B. Now, the total magnetic flux of each lateral winding entering small bands on the stator $\phi_{s.ax.k}$ and on the rotor $\phi_{r.ax.k}$ are calculated by (15) and (16) respectively.

$$\begin{aligned} \phi_{s.ax.k} &= \int_0^{\frac{\pi}{p}} B_{s.ax.k}(r = r_k, z = 0, \theta) r_k m d\theta \\ &= \sum_{n=2k+1} [M_n + N_n + \psi_n 4r_k^2 \mu_0 p^{-2} n^{-2}] m \quad (15) \\ \phi_{r.ax.k} &= \int_0^{\frac{\pi}{p}} B_{r.ax.k}(r = r_k, z = h_k, \theta) r_k m d\theta = \\ &\sum_{n=2k+1} \left[M_n e^{\frac{np}{2r_k} h_k} + N_n e^{-\frac{np}{2r_k} h_k} + \psi_n 4r_k^2 \mu_0 p^{-2} n^{-2} \right] m \quad (16) \end{aligned}$$

where ψ_n , M_n and N_n are defined by equations (A.4), (A.28), and (A.29), respectively, and also h_k is shown in Fig. 13(c).

3) *Magnetic Flux Density Calculation on the q -Axis Sections of the Stator and the Rotors*: In the mentioned critical operation point, Average magnetic flux densities on the q -axis of the trapezoid cuts of the stator $B_{sat.s}$ and the rectangular cross sections of the rotors $B_{sat.r}$ with thickness d_r and inner radius $r_{i.r}$ are defined by (17) and (18), respectively

$$B_{sat.s} = \frac{\phi_{T.g} + \phi_{s.in} + \phi_{s.out} + 2 \sum_{k=1}^i \phi_{s.ax.k}}{(d_i + d_o)(r_o - r_i) \times 0.5} \quad (17)$$

$$B_{sat.r} = \frac{(\phi_{T.g}/2) + \sum_{k=1}^i \phi_{r.ax.k}}{(d_r)(r_o - r_{i.r})} \quad (18)$$

C. Torque Calculations

Based on the proposed MEC, the Lorentz force law and the relationship of torque with force, the differential torque produced by each PM at radius r as $dT_{Mag.}(r)$ is calculated as follows:

$$dT_{Mag.}(r) = s_f J_{Cond.} r_o \theta_{eff} h_{Cop.o} r B_g(r) dr \quad (19)$$

where $J_{Cond.}$, θ_{eff} , and s_f are, respectively, the current density of each conductor, the effective air gap angle involved in torque production and the winding fill factor.

By considering the actuator pole number, the total output torque $T_{T.Mag.}$ is calculated by integrating (19) as follows:

$$\begin{aligned} T_{T.Mag.} &= 2p \int_{r_i}^{r_o} dT_{T.Mag.}(r) dr = G \left[\frac{(a_3 r)^2}{2} \right. \\ &\quad \left. + a_2^2 \ln(a_2 + a_3 r) - a_2 a_3 r \right] \Big|_{r: r_i \rightarrow r_o}, \quad p = 2, 4, \dots \end{aligned} \quad (20)$$

TABLE I
DATA OF CHOSEN INDUSTRIAL LATMS FOR COMPARISON WITH DAAT

Manufacturer	Aeroflex Co.	H2W Tech Co.
Type	Pancake Type	Radial Type
Model Number	TQ40W-16PKA	TMR-020-270-4V
Angular Excursion	120(deg.)	20(deg.)
Continuous Torque	0.847(N.m)	1.871(N.m)
Peak Torque	1.836(N.m)	5.6194(N.m)
Peak Power	200(watts)	71.53(watts)
Outer Diameter	10.16(Cm)	13.335(cm)
Height	3.81(Cm)	4.318(cm)

where

$$G = 2pJ_{\text{Cond.}} r_o \theta_{\text{eff.}} h_{\text{Cop.o}} s_f a_1 a_3^{-3} \quad (21)$$

where a_1 , a_2 , and a_3 have previously been defined by (8)–(10). Here, $\theta_{\text{eff.}}$ is defined as the average of PM angle $\theta_{\text{Mag.}}$ and θ_g . So, by considering the fringing coefficient, the relation of $\theta_{\text{Mag.}}$ and $\theta_{\text{eff.}}$ is given as

$$\theta_{\text{Mag.}} = 2s(1+s)^{-1}\theta_{\text{eff.}} \quad (22)$$

D. Loss Calculation

LATMs are fed by dc sources and their rotations are not continuous in static operation. Therefore, iron loss is completely neglected in this condition and the ohmic loss in the armature is the only source of the electromagnetic losses in LATMs. To calculate the ohmic loss in DAATs, the average length of each coil $l_{\text{Coil.Avg}}$ should first be calculated as:

$$l_{\text{Coil.Avg}} = d_i + d_o + 2\sqrt{\left(\frac{h_{\text{Cop.r}_i} - h_{\text{Cop.r}_o}}{2}\right)^2 + (r_o - r_i)^2} + \frac{\pi}{2}(h_{\text{Cop.r}_i} + h_{\text{Cop.r}_o}) \quad (23)$$

To produce a uniform torque profile, each PM should face similar armature winding distribution during the constant torque range. Hence, each winding angle σ relates to the constant torque angle γ by the following equation:

$$\sigma = \gamma + \theta_{\text{eff}} \quad (24)$$

Now, the total ohmic loss of DAAT is found as follows:

$$P_{\text{Loss.T}} = \rho_{T^\circ\text{C}} \times p \times s_f \times l_{\text{Coil.Avg}} \times r_o \times \sigma \times h_{\text{Cop.o}} \times J_{\text{Cond.}}^2 \quad (25)$$

where $\rho_{T^\circ\text{C}}$ is the conductor resistivity at T ($^\circ\text{C}$).

IV. SIMULATIONS AND COMPARISONS

In order to demonstrate the improved performances of the proposed structure in comparison with the previous industrial LATMs and the conventional DSAPM structure (DAAR), two optimum DAATs were designed and simulated by the authors. Their behaviors are studied in the current section. Designing was done by using the direct search method based on the derived equations in the previous sections to achieve maximum output capability with no saturation in short-time operations. Data for the industrial pancake type and the slotless radial LATM with distributed windings are summarized in Table I.

TABLE II
DESIGNED DAAT-20 TO BE COMPARED WITH THE RADIAL TYPE

$T_{T.\text{Mag.}}(\text{Peak}) = 5.62 \text{ Nm}$ & $\gamma = 20 \text{ deg.}$ & $\text{Height} = 35.3 \text{ mm}$			
r_o	65mm	d_o	12.4mm
r_i	24.5mm	d_i	6.1mm
l_m	5.2mm	d_r	3.4mm
θ_{Mag}	33.14deg.	$h_{\text{cop.}(r_o)}$	1.9mm
σ	54.98deg.	$P_{\text{Loss.T}}$	55.9watts
J	7A.mm-2	h_r	35.34mm

TABLE III
DESIGNED DAAT-120 TO BE COMPARED WITH THE PANCAKE TYPE

$T_{T.\text{Mag.}}(\text{Peak}) = 1.85 \text{ Nm}$ & $\gamma = 120 \text{ deg.}$ & $\text{Height} = 47.12 \text{ mm}$			
r_o	50mm	d_o	17.8mm
r_i	18.5mm	d_i	11.4mm
l_m	6.9mm	d_r	4.9mm
θ_{Mag}	46.2deg.	$h_{\text{cop.}(r_o)}$	1.86mm
σ	168.8deg.	$P_{\text{Loss.T}}$	81.6watts
J	9A.mm-2	h_r	47.1mm

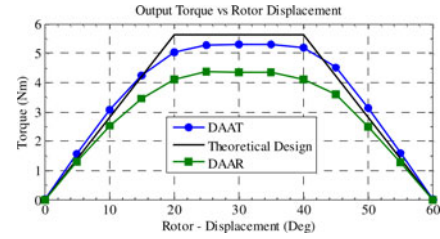


Fig. 7. Torque versus rotor displacement for actuators with a constant torque range of 20° .

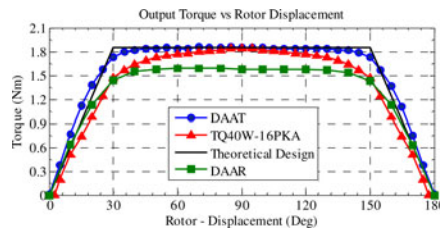


Fig. 8. Torque versus rotor displacement for actuators with a constant torque range of 120° .

A. Simulations

To validate the proposed design method and the derived equations, the authors used FE analysis on the designed geometries. The magnetic flux density distribution in the ferromagnetic parts of DAAT-20 and DAAT-120 with design parameters that are, respectively, summarized in Tables II and III, is shown in Fig. 6.

The output torque profiles versus rotor displacement obtained by the static FE analysis are plotted in Figs. 7 and 8. To support the applied torque on a nonferromagnetic shaft, the diameters of the shafts for DAAT-120 and DAAT-20 are assumed to be 1 and 2 cm, respectively.

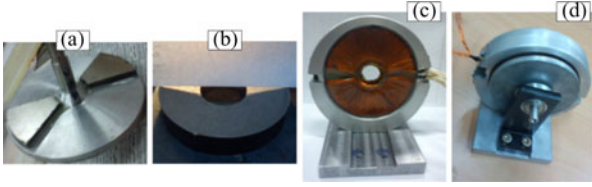


Fig. 9. Prototype of the DAAT-120: (a) rotor, (b) stator with the trapezoidal cross section, (c) wounded stator, and (d) the complete assembly.

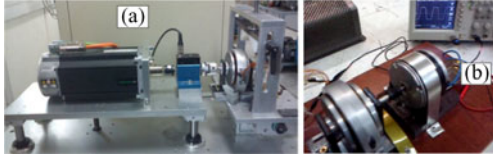


Fig. 10. Test setups: (a) torque versus angle measurement; (b) back-emf measurement.

B. Comparisons

1) *DAAT with DAAR (Conventional DSAPM)*: As it is observed in Fig. 7, DAAT-20 shows an improvement of 22.06% in output torque with a maximum of 5.31 N·m compared to DAAR with a maximum of 4.35 N·m. Input power and volume of both actuators are the same.

Fig. 8 shows a 17.02% improvement in output torque of DAAT-120 with a maximum of 1.849 N·m compared to the maximum value 1.58 N·m for DAAR. Input power and volume of both actuators are the same.

These improvements are achieved only by reshaping the stator from the conventional rectangular cross section to the proposed trapezoidal cross section.

2) *DAAT with the Slotless Radial Flux LATM*: As it can be deduced from Fig. 7 and also Tables I and II, the input power and the total volume of the DAAT-20, respectively, decreased by 17.46% and 21.85% compared to *TMR-020-270-4 V*. Theoretically, an output torque of 5.62 N·m was obtained for the proposed design, which compares well with the 5.31 N·m calculated by the FE analysis; an error of 5.52%.

3) *DAAT with the Pancake Type LATM*: The applied input power to the DAAT-120 significantly decreased by 59.2% in comparison with *TQ40 W-16PKA*, while increment of the total volume was 28.88%.

Furthermore, as it can be observed in Fig. 8, the slotless stator structure provides almost a uniform torque profile in the constant torque range with a maximum drop of 6.44%, while this drop in the slotted *TQ40 W-16PKA* structure is 20%.

V. PROTOTYPING

To confirm the validity and applicability of the proposed rotary actuator, DAAT-120 made out of CRGO-M5 ($B_{\text{sat}} = 1.7$ T) and grade N35 rare earth magnets ($H_c = 860$ kA/m and $\mu_m = 1.086\mu_0$) has been prototyped as in Fig. 9 with the parameters summarized in Table III.

For 80 W of input power, the output torque and the no-load back-emf were recorded by the setup instruments of Fig. 10 and plotted in Figs. 11 and 12, respectively. With 80 W of input power, the discrepancy between the 1.623 N·m measured output

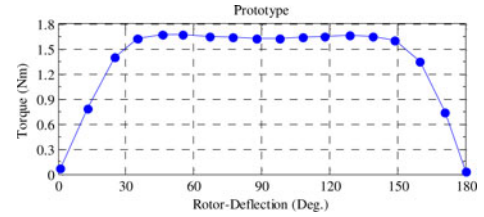


Fig. 11. Torque versus rotor deflection for the prototyped DAAT-120.

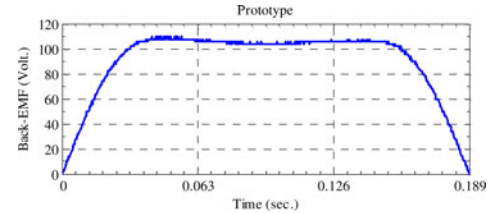


Fig. 12. Half-cycle waveform of measured back-EMF for the DAAT-120.

torque at 90° and the value obtained by the FE simulation was 12%.

VI. CONCLUSION

Based on the modification of the conventional DSAPM, the authors proposed, designed, and prototyped a new structure for LATMs. The electromagnetic capability notably increased with the implementation of the proposed trapezoidal cross section for stator with minimum air gaps. The governing set of equations for DAAT was derived, the armature reaction field calculated, and its characteristics validated by FEM and prototyping.

Comparisons with the latest industrial LATMs were done through the FEM and measurements, which illustrated:

- 1) torque increase by at least 17% for the same volume and input power compared to the conventional DSAPM;
- 2) noticeable decrease in input power by 17.4% and total volume by 21.8% in comparison with the slotless radial flux LATM; and
- 3) significant decrease in input power by 59% and an increase in total volume by 28.9% compared to the pancake LATM. Although DAAT cannot compete with the pancake type in torque to volume ratio, but its torque profile is almost completely smooth, which is not achieved by the pancake type LATMs.

APPENDIX A

A. Flux Passing Through the Inner and the Outer Walls of the Stator (Caused by the Axial Armature Current)

Magnetic vector potential is defined by the 2-D Maxwell's equations in the cylindrical coordinates. The ferromagnetic parts are assumed infinitely permeable.

1) *Solving Maxwell's Equations for the Inner Wall of the Stator*: By ignoring the end effects, magnetic vector potential A has zero value in all directions except along the z -axis as illustrated in Fig. 13(a). The Maxwell's equations in free space (medium I) and conductors (medium II) are as in (A.1) and

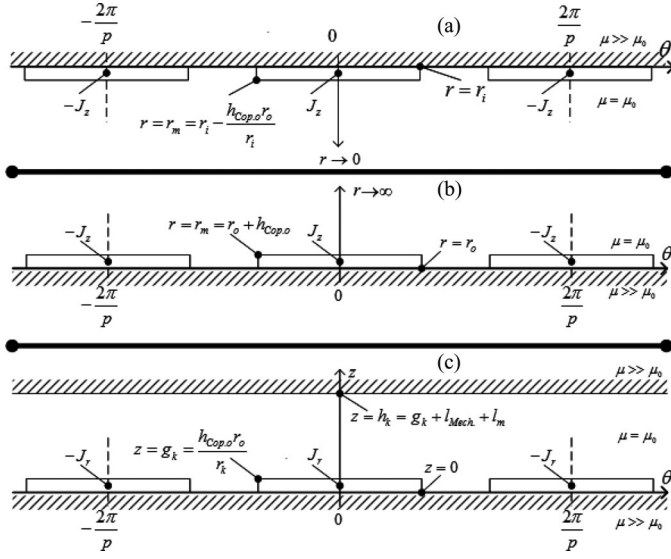


Fig. 13. Analyzed area: (a) outer side; (b) inner side; (c) lateral side.

(A.2).

$$\frac{\partial^2 A_{zI}}{\partial r^2} + \frac{1}{r} \frac{\partial A_{zI}}{\partial r} + \frac{1}{r^2} \frac{\partial^2 A_{zI}}{\partial \theta^2} = 0 \quad (A.1)$$

$$\frac{\partial^2 A_{zII}}{\partial r^2} + \frac{1}{r} \frac{\partial A_{zII}}{\partial r} + \frac{1}{r^2} \frac{\partial^2 A_{zII}}{\partial \theta^2} = -\mu_0 J_z \quad (A.2)$$

where J_z is the axial current density passing through the conductors that lean against the inner wall of the stator. The Fourier series expansion of this current density is

$$J_z(\theta) = \sum_{n=2k+1} \left[\psi_n \cos\left(\frac{np}{2}\theta\right) \right], \quad k = 0, 1, \dots \quad (A.3)$$

where,

$$\psi_n = 4J_{\text{Cond}} s_f \sin\left(\frac{np}{2}\alpha\right) (\pi n)^{-1}, \quad \alpha = \sigma/2 \quad (A.4)$$

The general solution of (A.1) and (A.2) is determined by (A.5) and (A.6), respectively [15]

$$A_{zI} = \sum_{n=2k+1} [C_n r^{\frac{np}{2}} + D_n r^{-\frac{np}{2}}] \cos\left(\frac{np}{2}\theta\right) \quad (A.5)$$

$$A_{zII} = \begin{cases} \sum_{n=2k+1} \left[(E_n r^{\frac{np}{2}} + F_n r^{-\frac{np}{2}}) + \psi_n \frac{-\mu_0}{(4 - (n^2 p^2 / 4))} r^2 \right] \cos\left(\frac{np}{2}\theta\right), & np \neq 4 \\ \sum_{n=2k+1} \left[(E_n r^{\frac{np}{2}} + F_n r^{-\frac{np}{2}}) + \psi_n \frac{-\mu_0}{4} r^2 \ln(r) \right] \times \cos\left(\frac{np}{2}\theta\right), & np = 4 \end{cases} \quad (A.6)$$

The following boundary conditions are applicable to the analyzed area [see Fig. 13(a)]:

$$\vec{B}_{II} \cdot \vec{a}_\theta |_{r=r_i} = 0 \quad (A.7)$$

$$\vec{B}_I \cdot \vec{a}_r |_{r=r_m} = \vec{B}_{II} \cdot \vec{a}_r |_{r=r_m} \quad (A.8)$$

$$\vec{H}_I \cdot \vec{a}_\theta |_{r=r_m} = \vec{H}_{II} \cdot \vec{a}_\theta |_{r=r_m} \quad (A.9)$$

Here, A , magnetic flux density B , and magnetic field strength H are related to each other as

$$\nabla \times A_z \vec{a}_z = B_r \vec{a}_r + B_\theta \vec{a}_\theta = \mu_0 \vec{H} \quad (A.10)$$

Now, C_n, D_n, E_n , and F_n are calculated by considering $r = 0$ in medium I, above boundary conditions, and (A.10) as the following:

$$D_n = 0 \quad (A.11)$$

$$F_n = \begin{cases} r_m^{2+\frac{np}{2}} \alpha_n \left(2 - \frac{np}{2}\right) (np)^{-1}, & np \neq 4 \\ r_m^{2+\frac{np}{2}} \alpha_n \left(2 \ln(r_m) + 1 - \frac{np}{2}\right) (np)^{-1}, & np = 4 \end{cases} \quad (A.12)$$

$$E_n = \begin{cases} \frac{\frac{np}{2} F_n r_i^{-\frac{np}{2}-1} - 2\alpha_n r_i}{\frac{np}{2} r_i^{\frac{np}{2}-1}}, & np \neq 4 \\ \frac{\frac{np}{2} F_n r_i^{-\frac{np}{2}-1} - \alpha_n r_i (2 \ln(r_i) + 1)}{\frac{np}{2} r_i^{\frac{np}{2}-1}}, & np = 4 \end{cases} \quad (A.13)$$

$$C_n = \begin{cases} F_n r_m^{np} + E_n + r_m^{2-(np/2)} \alpha_n, & np \neq 4 \\ F_n r_m^{np} + E_n + r_m^{2-(np/2)} \alpha_n \ln(r_m), & np = 4 \end{cases} \quad (A.14)$$

where

$$\alpha_n = \begin{cases} -\mu_0 \psi_n \left(4 - \frac{n^2 p^2}{4}\right)^{-1}, & np \neq 4 \\ \psi_n \frac{-\mu_0}{4}, & np = 4 \end{cases} \quad (A.15)$$

2) *Solving Maxwell's Equations for the Outer Wall of the Stator:* Governing equations of the analyzed area shown in Fig. 13(b) are similar to (A.1) through (A.10) by replacing C_n, D_n, E_n, F_n , and r_i with G_n, H_n, I_n, J_n , and r_o , respectively. By considering $r \rightarrow \infty$ in medium I, the previously mentioned boundary conditions and (A.10), we have

$$G_n = 0 \quad (A.16)$$

$$I_n = \begin{cases} -r_m^{2-(np/2)} \alpha_n \left(2 + \frac{np}{2}\right) (np)^{-1}, & np \neq 4 \\ -r_m^{2-(np/2)} \alpha_n \left(2 \ln(r_m) + 1 + \frac{np}{2}\right) (np)^{-1}, & np = 4 \end{cases} \quad (A.17)$$

$$J_n = \begin{cases} \frac{(np/2) I_n r_o^{(np/2)-1} + 2\alpha_n r_o}{(np/2) r_o^{-(np/2)-1}}, & np \neq 4 \\ \frac{(np/2) I_n r_o^{(np/2)-1} + \alpha_n r_o (2 \ln(r_o) + 1)}{(np/2) r_o^{-(np/2)-1}}, & np = 4 \end{cases} \quad (A.18)$$

$$H_n = \begin{cases} I_n r_m^{np} + J_n + r_m^{2+(np/2)} \alpha_n, & np \neq 4 \\ I_n r_m^{np} + J_n + r_m^{2+(np/2)} \alpha_n \ln(r_m), & np = 4 \end{cases} \quad (A.19)$$

APPENDIX B

A. Flux Passing Through the Lateral Surfaces of the Stator (Caused by the Radial Armature Current)

In this section, the magnetic vector potential is defined by the Maxwell's equations in the angular cutting of the actuator in the cylindrical coordinates. The ferromagnetic parts are assumed infinitely permeable. The analyzed area corresponding to the one lateral side of the actuator for $r = r_k$ is shown in Fig. 13(c).

By ignoring the end effects, the magnetic vector potential A has zero value in all directions except along the r -axis. Static Maxwell's equations in the free space (medium I) and the conductors (medium II) are as in (A.20) and (A.21)

$$\frac{1}{r^2} \frac{\partial^2 A_{rI}}{\partial \theta^2} + \frac{\partial^2 A_{rI}}{\partial z^2} = 0 \quad (\text{A.20})$$

$$\frac{1}{r^2} \frac{\partial^2 A_{rII}}{\partial \theta^2} + \frac{\partial^2 A_{rII}}{\partial z^2} = -\mu_0 J_r \quad (\text{A.21})$$

where J_r is radial current density through conductors that can be obtained by substituting J_r for J_z in (A.3) and (A.4).

By using the separation of variable technique, the general solutions of (A.20) and (A.21) are

$$A_{rI} = \sum_{n=2k+1} [K_n e^{\frac{np}{2r}z} + L_n e^{-\frac{np}{2r}z}] \cos\left(\frac{np}{2}\theta\right) \quad (\text{A.22})$$

$$A_{rII} = \sum_{n=2k+1} \left[(M_n e^{\frac{np}{2r}z} + N_n e^{-\frac{np}{2r}z}) + \frac{4\psi_n \mu_0 r^2}{p^2 n^2} \right] \times \cos\left(\frac{np}{2}\theta\right). \quad (\text{A.23})$$

The following boundary conditions are applicable to the analyzed area [see Fig. 13(c)]:

$$\vec{B}_I \cdot \vec{a}_\theta|_{z=0} = \vec{B}_{II} \cdot \vec{a}_\theta|_{z=h_k} = 0 \quad (\text{A.24})$$

$$\vec{B}_I \cdot \vec{a}_z|_{z=g_k} = \vec{B}_{II} \cdot \vec{a}_z|_{z=g_k} \quad (\text{A.25})$$

$$\vec{H}_I \cdot \vec{a}_\theta|_{z=g_k} = \vec{H}_{II} \cdot \vec{a}_\theta|_{z=g_k} \quad (\text{A.26})$$

Here, A_r , magnetic flux density B , and magnetic field strength H are related to each other by (A.27)

$$\nabla \times A_r \vec{a}_r = B_\theta \vec{a}_\theta + B_z \vec{a}_z = \mu_0 \vec{H} \quad (\text{A.27})$$

Finally, K_n, L_n, M_n , and N_n are calculated by using the expressed boundary conditions and (A.27) as the following:

$$M_n = \frac{4\psi_n \mu_0 r^2 / p^2 n^2}{\frac{[1 - e^{\frac{np}{2r}g_k}][e^{\frac{np}{2r}g_k} + e^{-\frac{np}{2r}g_k} e^{\frac{np}{2r}h_k}] - [e^{\frac{np}{2r}h_k} - e^{-\frac{np}{2r}g_k}]}{[e^{\frac{np}{2r}h_k} - e^{-\frac{np}{2r}g_k}]} - [e^{\frac{np}{2r}g_k} + e^{-\frac{np}{2r}g_k}]} \quad (\text{A.28})$$

$$N_n = M_n \quad (\text{A.29})$$

$$K_n = \frac{[1 - e^{\frac{np}{2r}g_k}]}{[e^{\frac{np}{2r}h_k} - e^{-\frac{np}{2r}g_k}]} M_n \quad (\text{A.30})$$

$$L_n = K_n e^{\frac{np}{2r}h_k}. \quad (\text{A.31})$$

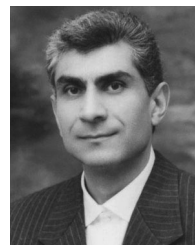
REFERENCES

- [1] T. Ching-Chih, L. Shui-Chun, H. Hsu-Chih, and C. Yu-Ming, "Design and control of a brushless DC limited-angle torque motor with its application to fuel control of small-scale gas turbine engines," *Elsevier J. Mechatron.*, vol. 19, pp. 29–41, Feb. 2009.
- [2] H. R. Bolton and Y. Shakweh, "Performance prediction of Law's relay actuator," in *Proc. IEE*, vol. 137, no. 1, pp. 1–13, Jan. 1990.
- [3] Y. Qiu, D. J. Perreault, J. G. Kassakian, and T. A. Keim, "A custom-designed limited-angle actuator for an electromechanical engine valve drive part I: Conceptual design," in *Proc. 5th IET Int. PEMD Conf.*, 2010, pp. 1–6.
- [4] F. S. Ahmed, S. Laghrouche, and M. El Bagdouri, "Analysis, modeling, identification and control of pancake DC torque motors: Application to automobile air path actuators," *Elsevier J. Mechatron.*, vol. 22, pp. 195–212, Mar. 2012.
- [5] T. T. Overboom, J. W. Jansen, E. A. Lomonova, and F. Tacken, "Design and optimization of a rotary actuator for a two degree-of-freedom $z\phi$ -module," *IEEE Trans. Ind. Appl.*, vol. 46, no. 6, pp. 2401–2409, Nov./Dec. 2010.
- [6] W. Guo, S. Weerasooriya, T. B. Goh, Q. H. Li, C. Bi, and K. T. Chang, "Dual stages actuators for high density rotating memory devices," *IEEE Trans. Magn.*, vol. 34, no. 2, pp. 450–455, Mar. 1998.
- [7] C. Dawson and H. R. Bolton, "Design of a class of wide-angle limited-rotation rotary actuators," *IEE Proc.*, vol. 126, no. 4, pp. 345–350, 1979.
- [8] C. Dawson and H. R. Bolton, "Performance prediction of a wide-angle limited-motion rotary actuator," *IEE Proc.*, vol. 125, no. 9, pp. 895–898, 1978.
- [9] C. Dawson and H. R. Bolton, "Limited motion rotary actuators of the toroidal-stator, permanent-magnet rotor type," *IEE Proc. Electr. Power Appl.*, vol. 129, pp. 190–198, 1982.
- [10] H. Nakazato, "Rotary voice coil motor with a flat coil," U.S. Patent 5 122 702, Jan. 16, 1992.
- [11] P. C. Chou, Y. C. Lin, and S. Cheng, "A novel seesaw swivel actuator design and fabrication," *IEEE Trans. Magn.*, vol. 46, no. 7, pp. 2603–2610, Jul. 2010.
- [12] D. B. Mohler, "Permanent magnet brushless actuator," U.S. Patent 5 337 030, Aug. 9, 1994.
- [13] E. A. Mendrela, R. Beniak, and R. Wrobel, "Influence of stator structure on electromechanical parameters of torus-type brushless DC motor," *IEEE Trans. Energy Convers.*, vol. 18, no. 2, pp. 231–237, Jun. 2003.
- [14] K. Sitapati and R. Krishnan, "Performance comparisons of radial and axial field, permanent-magnet, brushless machines," *IEEE Trans. Ind. Appl.*, vol. 37, no. 5, pp. 1219–1226, Sep./Oct. 2001.
- [15] O. Wallmark and P. Kjellqvist, "Analysis of a low-cost air-gap winding, for permanent magnet synchronous motors," *IEEE Trans. Energy Convers.*, vol. 24, no. 4, pp. 841–847, Dec. 2009.



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